RISKLENS

AI-powered financial event risk intelligence

Turn unstructured financial news and social signals into interpretable risk signals, then explore potential portfolio impact through scenario-based stress testing.

S&P Global & CRISIL Campus Hackathon 2026 • Prototype presentation

01 | The problem

Financial risk information arrives as unstructured text across multiple channels. Manually interpreting each event and translating it into portfolio exposure can be slow and inconsistent.

Our goal

Extract meaningful events, assess sentiment and impact consistently, then connect those signals to portfolio-level scenario analysis.

02 | Solution overview

RiskLens connects text ingestion and data-quality checks with a risk engine, then makes structured signals available through an API and dashboard.

Pipeline

1. Ingest financial news and social-media text → 2. Validate, normalize and extract entities → 3. Estimate sentiment, event type, impact and confidence → 4. Emit risk signal → 5. Apply portfolio stress scenarios.

Implementation: React dashboard • FastAPI REST API • MongoDB-backed persistence when configured • YAML-configured scenarios

03 | Risk signal engine

A submitted text record is transformed into a structured risk signal for downstream services.

Key outputs

Sentiment score (-1 to +1), event classification, impact score (1–10), confidence, matched keywords and provenance.

Quality controls

Input validation, normalization, entity extraction and data-quality checks help identify malformed or inconsistent inputs. Source credibility and context still require human review.

04 | Portfolio stress testing

The downstream module applies preset or user-defined percentage shocks to portfolio asset classes and estimates the resulting change in value.

Outputs

Portfolio value before/after, absolute and percentage loss, and asset-level impact breakdown. Scenarios are configurable and results are explainable through explicit shocks.

Boundary

These are simplified scenario estimates—not forecasts, VaR, or investment advice.

05 | Live demo flow

1. Add synthetic portfolio holdings → 2. Submit a financial event → 3. Inspect sentiment, event type and impact → 4. Run a preset/custom stress test → 5. Review portfolio and asset-level losses.

Current prototype boundaries

Demo entities, event text, holdings, prices and shocks are synthetic. Outputs depend on implemented rules/models and input quality. This is not a production risk system and does not use live market feeds.

06 | Validation & next steps

Run automated backend tests and build the frontend before submission. Demonstrate API health, analysis output, portfolio CRUD and stress-test results.

Roadmap

Benchmark scoring on labelled data; add source-reliability signals and audit trails; calibrate scenarios with domain experts; harden monitoring and deployment.

All examples are fictional. Stress outputs are simplified assumptions, not forecasts or investment advice.